

Building a Reproducible Scholarly Search Toolkit

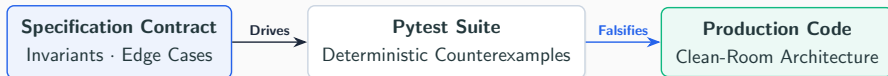
Episode 1: Reading the Scientific Contract

Nexus Scholar Suite

The Specification Contract

Design by Contract

Scientific software must not have ambiguous behaviors. We define explicit contracts: inputs, outputs, mathematical invariants, and failure modes.



Key Contract Invariants

Identifier Invariants

- Case-insensitive DOI matching.
- Prefix stripping (doi:, URLs).
- Nullable empty strings.

Temporal Invariants

- Explicit UTC timestamps.
- Year bounded ($1900 \leq y \leq 2100$).
- Non-zero publication dates.